LIFTING A SMOOTH PRESENTATION OF Y THROUGH F TO A SMOOTH PRESENTATION OF X

THE STACKS PROJECT AUTHORS

Editorial note. No omission markers in-span. The two transfer inputs are not carded ('Proof omitted.' each) and are declared external.

Allowed standard prerequisites (editorial).

- Algebraicity of 2-fibre products of algebraic stacks. lemma-2-fibre-product-general (carded L26685). The 2-fibre product of the presentations is an algebraic stack, hence equivalent to S_F.
- Base-change property transfer. lemma-base-change-representable-transformations-property (same chapter, NOT carded, declared external). S_F -> (Sch/W) is surjective smooth as a base change of y.
- Composition property transfer. lemma-composition-representable-transformations-property (same chapter, NOT carded, declared external). S_F -> X is surjective smooth; also the final etale composition.

Source reference. Original lemma 045C: sources/stacks-algebraic/pinned-original-source.tex, source lines 1180–1180. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Source reference. Original lemma 045B: sources/stacks-algebraic/pinned-original-source.tex, source lines 1159–1159. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Lemma 0.1. *Let S be a scheme contained in Sch_{fppf}. Let $f : \mathcal{X} \to \mathcal{Y}$ be a 1-morphism of algebraic stacks over S. Let $V \in \mathrm{Ob}((Sch/S)_{fppf})$. Let $y : (Sch/V)_{fppf} \to \mathcal{Y}$ be surjective and smooth. Then there exists an object $U \in \mathrm{Ob}((Sch/S)_{fppf})$ and a 2-commutative diagram*

$$\begin{array}{ccc} (Sch/U)_{fppf} & \xrightarrow{a} & (Sch/V)_{fppf} \\ {\scriptstyle x}\downarrow & & \downarrow{\scriptstyle y} \\ \mathcal{X} & \xrightarrow{f} & \mathcal{Y} \end{array}$$

with x surjective and smooth.

Proof. First choose $W \in \mathrm{Ob}((Sch/S)_{fppf})$ and a surjective smooth 1-morphism $z : (Sch/W)_{fppf} \to \mathcal{X}$. As $\mathcal{Y}$ is an algebraic stack we may choose an equivalence

$$j : \mathcal{S}_F \longrightarrow (Sch/W)_{fppf} \times_{f \circ z, \mathcal{Y}, y} (Sch/V)_{fppf}$$

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where F is an algebraic space. By Lemma 045C the morphism $\mathcal{S}_F \to (Sch/W)_{fppf}$ is surjective and smooth as a base change of y. Hence by Lemma 045B we see that $\mathcal{S}_F \to \mathcal{X}$ is surjective and smooth. Choose an object $U \in \mathrm{Ob}((Sch/S)_{fppf})$ and a surjective étale morphism $U \to F$. Then applying Lemma 045B once more we obtain the desired properties. $\square$